

Patent Submission Draft

Context-Aware Interactive Bezel for Storage Appliance State Visualization and Screenless User Interaction

Prepared for invention disclosure and attorney review

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1. Candidate Patent Titles

- Context-Aware Interactive Bezel for Storage Appliance Monitoring and Screenless User Interaction
- System and Method for Context-Aware Storage Appliance State Visualization Through an Interactive Bezel
- Adaptive Storage Appliance Bezel for Operational State Visualization, Alerting, and Local User Interaction
- AI-Agent Driven Bezel Interface for Enterprise Storage System State Visualization and User Interaction
- Contextual Human-Machine Interface Bezel for Monitoring and Managing Enterprise Storage Infrastructure

2. Invention Summary

The invention relates to an enterprise storage appliance bezel incorporating an addressable LED light strip, local sensing, and optional user interaction mechanisms. The bezel is driven by an AI agent or policy engine that monitors internal storage-array telemetry and external environmental signals, interprets operational context, and renders color, pattern, brightness, sequencing, or animation states around the appliance bezel to communicate system status, risk, activity, and recommended action without requiring a terminal, management console, or attached display.

The bezel may also operate as a localized human-machine interface. An authorized user may initiate a query or interaction through voice, gesture, proximity, button, mobile-device pairing, service badge, or another local input. The system can respond through LED state changes, pattern sequences, audible output, mobile push response, or integration with a service-management system. Example queries include capacity utilization, projected capacity exhaustion, detected data movement, cyber threat state, recovery posture, power or cooling risk, and system identity within a multi-array environment.

3. Short Abstract

A storage appliance bezel includes an addressable illumination element and a controller communicatively coupled to a storage operating environment. An AI agent receives internal telemetry and external environmental signals, determines an operational state of the storage appliance or managed environment, and controls the illumination element to display a corresponding visual condition. The bezel can also receive local user input and initiate operational queries or workflows without requiring a terminal or display. The invention enables contextual, screenless monitoring and interaction for enterprise storage infrastructure.

4. Technical Field

The invention relates generally to enterprise storage systems, physical infrastructure interfaces, human-machine interaction, telemetry-driven visualization, AI-assisted infrastructure operations, cyber-resilience monitoring, and ambient status indication for data center appliances.

5. Background and Problem Statement

Enterprise storage systems typically communicate operating status through remote management consoles, software dashboards, event logs, audible alarms, or limited front-panel indicators. These mechanisms often require an administrator to access a terminal, browser, service interface, or monitoring system before understanding the current state of the appliance or the data environment it supports.

Modern storage appliances increasingly manage complex operational states, including cyber-threat indicators, anomalous data access, potential data exfiltration, capacity growth, replication, tiering, high I/O patterns, sustainability conditions, cooling risk, power draw, workload locality, and multi-system coordination across fleets. Traditional status indicators do not convey this multidimensional context effectively at the physical appliance itself.

A need exists for a contextual, physical, and optionally interactive interface integrated into the storage appliance bezel that can translate storage telemetry, environmental data, AI-derived insights, and policy decisions into visible and actionable signals at the device location.

6. Core Inventive Concepts

Concept	Description	Potential Patentable Distinction
Telemetry-driven LED bezel	An addressable LED strip around or within a storage appliance bezel displays state-specific colors, brightness, patterns, animations, or sequences.	Converts both storage-array telemetry and data-environment states into physical visual signals at the appliance.
AI-agent state interpretation	An AI agent, policy engine, or rules engine evaluates internal and external signals and determines a context-aware bezel state.	Moves beyond static fault LEDs toward contextual incident, risk, workload, and environment visualization.
Screenless local interaction	Authorized users can query the storage system locally without a terminal or screen through voice, gesture, proximity, service badge, button, or mobile interface.	Turns the bezel from a passive indicator into an interactive infrastructure interface.
Internal and external signal fusion	Signals may include storage health, capacity trends, I/O patterns, data movement, cyber state, data exfiltration indications, decibel levels, power, cooling, network state, and proximity to peer systems.	Combines appliance operating data with environmental and fleet context.
Policy-based visual grammar	Color, animation, intensity, and segmentation can be mapped to severity, confidence, workload class, capacity state, resiliency posture, or cyber event type.	Provides a structured visual language for enterprise infrastructure status.
Fleet-aware physical identification	The bezel can identify an appliance within a rack or fleet, communicate relationship to adjacent systems, or guide service	Enables multi-system visualization and service guidance in dense data center environments.

	personnel to the correct unit.	
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7. Example System Components

1. Storage appliance, such as an IBM FlashSystem array or other enterprise storage system.
2. Bezel assembly mechanically coupled to a chassis front face.
3. Addressable LED strip, light pipe, segmented illumination channel, or distributed lighting element integrated around the bezel perimeter or across selected bezel regions.
4. Bezel controller or microcontroller coupled to the storage appliance management plane, service processor, array controller, or external management service.
5. AI agent, inference service, policy engine, or event correlation service that evaluates storage telemetry and external signals.
6. Internal telemetry sources including capacity, utilization, latency, I/O activity, cache state, replication state, rebuild state, firmware state, cyber indicators, encryption anomalies, data movement, and health events.
7. External signal sources including ambient sound, power draw, cooling condition, temperature, airflow, network state, rack proximity, peer appliance state, service-ticket status, and data center environmental sensors.
8. Local input system such as microphone, capacitive touch, button, QR/NFC service tag, badge reader, proximity sensor, camera, mobile pairing, or gesture sensor.
9. Output system including LED state changes, animation, audio feedback, mobile-device response, service workflow creation, or management-console event generation.

8. Example Operating Modes

Mode	Example Trigger	Example Bezel Behavior
Steady-state operations	Normal I/O and no active risk state	Soft steady illumination, low brightness, color aligned to system identity or policy.
Capacity trending	Capacity utilization crosses threshold or projected exhaustion within defined window	Progressive fill pattern or segmented amber state indicating current and projected capacity pressure.
Data movement	Replication, migration, rebuild, tiering, or workload relocation in progress	Directional animation indicating movement, synchronization, or background activity.
Cyber threat	Ransomware signal, anomalous encryption, abnormal access pattern, or exfiltration indicator	High-urgency color state and pulsing pattern, optionally escalating based on confidence and blast radius.
Recovery posture	Safeguarded copy status, recovery validation result, or clean-copy attestation	Distinct secure/recovery-ready color or pattern, with alert state if protection gaps appear.
Environmental risk	High temperature, cooling concern, abnormal power draw, excessive sound level, or network degradation	Localized segment illumination associated with the affected risk category.
Service identification	Technician requests location of unit from console or mobile device	Unique beacon pattern identifying the selected array in rack or fleet.
Local query	Authorized user asks capacity utilization or projected exhaustion	Bezel displays response through proportional color/segment pattern and optionally returns detailed response to paired device.

9. Detailed Description of Preferred Embodiments

9.1 Bezel Illumination Architecture

In one embodiment, a front bezel of a storage appliance includes an addressable LED strip arranged around an outer perimeter of the bezel. The strip may be continuous, segmented, embedded behind a light pipe, or positioned behind translucent or semi-translucent bezel materials. The illumination element may support RGB, RGBW, brightness control, animation sequencing, and segmented addressing. The bezel controller may receive commands from the storage appliance, a management appliance, cloud service, orchestration platform, or an AI agent.

9.2 AI Agent and Policy Engine

An AI agent monitors internal and external signals and determines a contextual state. The AI agent may classify operating conditions, forecast future states, correlate telemetry across multiple subsystems, or assign confidence and severity levels. In some embodiments, a policy engine constrains the AI agent by mapping specific event classes, confidence thresholds, and risk levels to permitted bezel behaviors. For example, a high-confidence cyber threat may override a low-priority capacity indicator, while a service-location command may run only after authentication.

9.3 Internal Telemetry Inputs

Internal telemetry may include storage pool utilization, data reduction performance, compression efficiency, drive health, adapter state, controller state, replication lag, rebuild activity, host connectivity, I/O intensity, latency distributions, workload identity, cache behavior, copy-services state, Safeguarded Copy state, anomaly detection outputs, encryption activity, data movement operations, and firmware or lifecycle events.

9.4 External and Environmental Inputs

External signals may include rack power draw, thermal conditions, cooling risk, ambient decibel level, vibration, airflow, proximity to peer storage systems, rack location, network fabric status, service-management incident state, facilities events, and physical access events. The system may combine external and internal signals to differentiate between appliance-specific risk and data-center environmental risk.

9.5 Visual Grammar and Prioritization

A visual grammar may define how colors, animation patterns, brightness, pulse rates, and bezel segments correspond to states. For example, blue may indicate normal operations, green may indicate recovery-ready status, amber may indicate capacity or environmental caution, red may indicate cyber or failure risk, purple may indicate AI-related data movement, and white beaconing may identify a specific system. The exact color mapping is configurable and may conform to customer accessibility needs, dark data center conditions, or enterprise policy.

9.6 Screenless User Interaction

The bezel may receive local input. An authorized user may ask a question, press a portion of the bezel, scan a service tag, present a badge, or pair a mobile device. The system may authenticate the user and allow query-only functions, advisory functions, or supervised workflow initiation. For example, a field technician may ask, “What is the current capacity utilization?” or “Which array is reporting anomalous encryption?” The system may provide an immediate bezel response and send a detailed explanation to an authorized mobile or management interface.

9.7 Multi-Array and Fleet Coordination

In a fleet embodiment, multiple storage appliances coordinate visual states. Adjacent arrays may show relationship, replication direction, capacity balancing, service grouping, cluster membership, or data movement. A management service may direct one or more bezels to illuminate in coordinated patterns to guide service personnel or visualize distributed operations across a rack, row, or data center.

9.8 Security and Governance

Because the bezel may expose operational state, the system may restrict certain states, query responses, or interaction types based on policy. Cyber events may be abstracted visually without exposing sensitive details to unauthorized observers. User interaction may require authentication, role-based access control, proximity constraints, service-ticket validation, or secure pairing. All interactions may be logged for auditability.

10. Example Methods

Method A: Contextual State Visualization

10. Receive internal telemetry from a storage appliance operating environment.
11. Receive one or more external environmental, rack, network, or peer-system signals.
12. Analyze the received signals using an AI agent, rules engine, or policy-driven event correlation process.
13. Determine a contextual state selected from capacity, data movement, cyber threat, recovery, health, environmental, service, or fleet coordination states.
14. Map the contextual state to a visual instruction including color, intensity, segment, animation, duration, or priority.
15. Transmit the visual instruction to the bezel controller.
16. Render the contextual state through the addressable bezel illumination element.

Method B: Screenless Local Query

17. Detect a local user interaction at or near the storage appliance bezel.
18. Authenticate or authorize the interaction using proximity, credential, service state, mobile pairing, or policy.
19. Receive a query related to storage state, capacity, cyber posture, data movement, environmental risk, or system identity.
20. Retrieve relevant telemetry and answer the query using the AI agent or management service.
21. Output a summarized response through the bezel illumination element and optionally transmit a detailed response to a paired device or management console.
22. Record the interaction and response in an audit log.

11. Preliminary Claim Concepts for Counsel Review

The following are illustrative claim concepts only. Patent counsel should rewrite, broaden, narrow, and structure claims based on prior-art review, filing strategy, and desired claim scope.

Claim Set A: System Claim

23. A storage appliance system comprising: a storage appliance chassis; a bezel coupled to the storage appliance chassis; an addressable illumination element integrated with the bezel; a controller coupled to the addressable illumination element; and an agentic control service configured to receive telemetry from a storage operating environment, determine a contextual operating state, and cause the controller to render the contextual operating state using the addressable illumination element.
24. The system of claim 1, wherein the telemetry includes at least one of capacity utilization, projected capacity consumption, data movement activity, storage health, I/O pattern, replication state, cyber threat indicator, encryption anomaly, or data exfiltration indicator.
25. The system of claim 1, wherein the agentic control service further receives at least one external signal selected from ambient sound, power consumption, cooling condition, temperature, network state, proximity to another storage appliance, rack location, or facilities telemetry.

26. The system of claim 1, wherein the contextual operating state is rendered as at least one of color, brightness, pulse rate, segmented illumination, directional animation, or beaconing pattern.
27. The system of claim 1, further comprising a local input interface configured to receive a screenless user query associated with the storage appliance.
28. The system of claim 5, wherein the agentic control service generates a response to the screenless user query and causes the addressable illumination element to render a visual response.
29. The system of claim 5, wherein the local input interface includes at least one of a microphone, button, capacitive touch sensor, proximity sensor, mobile-device pairing interface, QR code, NFC interface, badge reader, or gesture sensor.
30. The system of claim 1, wherein the agentic control service applies a policy to prioritize cyber threat states over non-critical operational states.
31. The system of claim 1, wherein a plurality of storage appliances coordinate respective bezel illumination elements to display a fleet state or inter-appliance relationship.
32. The system of claim 1, wherein a detected data exfiltration risk causes the addressable illumination element to render a distinct visual state different from a general hardware fault state.

Claim Set B: Method Claim

33. A computer-implemented method comprising receiving telemetry from a storage appliance; receiving an environmental or fleet signal associated with the storage appliance; determining, using an AI agent or policy engine, a contextual state of the storage appliance or related data environment; mapping the contextual state to a bezel illumination instruction; and causing an addressable illumination element integrated with a storage appliance bezel to display the contextual state.
34. The method of claim 1, further comprising receiving a local user query without use of a terminal or display and rendering a response through the addressable illumination element.
35. The method of claim 1, wherein determining the contextual state includes classifying an event as a cyber threat, data movement operation, capacity trend, recovery posture, environmental risk, or service identification state.
36. The method of claim 1, further comprising authenticating a user before rendering sensitive operational information through the storage appliance bezel.
37. The method of claim 1, further comprising coordinating illumination patterns across a plurality of storage appliance bezels to indicate a replication relationship, service group, cluster membership, physical location, or data movement path.

12. Suggested Drawing Figures

Figure	Description
FIG. 1	Front perspective view of a storage appliance with an interactive LED bezel surrounding the front face.
FIG. 2	Block diagram showing internal storage telemetry, external environmental signals, AI agent, policy engine, bezel controller, and illumination element.
FIG. 3	Flowchart for telemetry ingestion, contextual state determination, policy mapping, and bezel-state rendering.
FIG. 4	Flowchart for local user query, authentication, answer generation, visual response, and audit logging.
FIG. 5	Example visual grammar showing colors, patterns, severity states, and priority override behavior.
FIG. 6	Multi-array fleet embodiment showing coordinated bezel illumination across storage systems within a rack.
FIG. 7	Example cyber-threat state transition from normal operation to anomalous encryption detection to recovery-ready response.

13. Differentiating Features to Emphasize During IP Review

- The bezel is not merely decorative and is not limited to static hardware status. It is driven by contextual interpretation of storage, data, cyber, environmental, and fleet states.
- The system turns an enterprise storage array into a physical endpoint for AI-assisted operational awareness.
- The AI agent can reason over state changes and translate them into a controlled visual grammar.
- The invention can enable screenless interaction with storage infrastructure at the rack location.
- The system can fuse internal operating telemetry with external environmental and proximity signals.
- The invention can coordinate visual state across multiple storage appliances to represent relationships, risks, or workflows.
- The invention can provide differentiated visual states for cyber threats, data exfiltration, capacity projections, recovery posture, and data movement rather than only device fault states.

14. Open Questions for Inventor and IBM IP Counsel

- What physical input mechanisms are actually contemplated or prototyped: microphone, touch, button, NFC, mobile pairing, badge, gesture, camera, or other?
- Is the AI agent fully on-array, in Storage Insights, in a cloud service, or distributed across appliance and management plane?
- What system states should be treated as non-public or sensitive if visually exposed in a data center?
- What color and animation states have already been designed, tested, or documented?
- Does the bezel include sensors, or does it only receive external sensor signals from the rack, data center, or management plane?
- Can the bezel initiate workflows, or is interaction limited to queries and status responses?
- Does the design support accessibility alternatives for color-blind operators, including patterns, segments, pulse rates, or companion notifications?
- Can multiple bezels coordinate to visualize data movement or replication relationships across arrays?
- Was any part publicly disclosed, demonstrated, shown at an event, or shared externally? If yes, record the dates and audiences immediately.
- Are there invention records, photographs, CAD files, prototype notes, Jira items, design reviews, or emails that establish conception and reduction-to-practice timing?

15. Potential Commercial and Operational Benefits

- Improves at-a-glance operational awareness in data centers and service environments.
- Helps administrators and field technicians identify system state without a management console.
- Provides a new physical interface for AI-assisted infrastructure operations.
- Supports cyber-resilience messaging by making threat, recovery, and protection states visible at the storage appliance.
- May reduce service time by guiding technicians directly to the affected system and visually communicating relevant state.
- Creates a differentiated industrial design and user experience for IBM FlashSystem and potentially other IBM infrastructure appliances.
- Enables future expansion into fleet-level, agentic, or autonomous operations visualization.

16. Filing Package Checklist

- Identify all inventors and their specific contributions.
- Collect design sketches, prototype photos, CAD drawings, firmware notes, AI-agent architecture diagrams, and telemetry mapping documents.
- Document conception date, design review dates, prototype dates, and any public/private disclosures.
- Prepare figure drawings or block diagrams based on the suggested figures above.
- Ask IBM IP counsel to perform prior-art search and determine utility, design, or combined filing strategy.
- Review whether separate filings should cover bezel industrial design, telemetry-to-visual-state methods, screenless interaction, AI-agent integration, and fleet coordination.
- Confirm whether the invention should be filed as a provisional application first or directly as a nonprovisional application based on IBM IP strategy.

17. Plain-English One-Sentence Invention Statement

The invention turns the storage appliance bezel into an AI-driven, context-aware visual and interactive interface that communicates storage, cyber, capacity, environmental, and fleet states at the physical array without requiring an operator to open a console or attach a display.